U6

① $f(x) = x^2 - 2x + 3 \quad \eta = 0{,}2 \quad f'(x) = 2x - 2$

t	x	f(x)	f'(x)
0	-2,0	11	-6
1	-0,8	5,24	-3,6
2	-0,08	3,1664	-2,16
3	0,352	2,41199	-1,296
4	0,6112	2,1512	-0,7776

$x_{t+1} = x_t - \eta f'(x_t)$

$f'(x_0) = 2(-2) - 2 = -6$

$f(x_0) = (-2)^2 - 2(-2) + 3 = 11$

$x_1 = -2{,}0 - 0{,}2 \cdot (-6) = -0{,}8$

$f'(x_1) = 2(-0{,}8) - 2 = -1{,}6 - 2 = -3{,}6$

$f(x_1) = (-0{,}8)^2 - 2(-0{,}8) + 3 = 0{,}64 + 3 = 5{,}24$

$x_2 = -0{,}8 - 0{,}2 \cdot (-3{,}6) = -0{,}8 + 0{,}72 = -0{,}08$

$f'(x_2) = 2(-0{,}08) - 2 = -2{,}16$

$f(x_2) = (-0{,}08)^2 - 2(-0{,}08) + 3 = 0{,}0064 + 0{,}16 + 3 = 3{,}1664$

$x_3 = -0{,}08 - 0{,}2 \cdot (-2{,}16) = -0{,}08 + 0{,}432 = 0{,}352$

$f'(x_3) = 2 \cdot (0{,}352) - 2 = 0{,}704 - 2 = -1{,}296$

$f(x_3) = 0{,}352^2 - 2(0{,}352) + 3 = 0{,}123904 - 0{,}704 + 3 = 2{,}419904$

$x_4 = 0{,}352 - 0{,}2 \cdot (-1{,}296) = 0{,}352 + 0{,}2592 = 0{,}6112$

$f'(x_4) = 2(0{,}6112) - 2 = 1{,}2224 - 2 = -0{,}7776$

$f(x_4) = 0{,}6112^2 - 2(0{,}6112) + 3 = 0{,}3736 - 1{,}2224 + 3 = 2{,}1512$

② a) if the x has 2 dimensions

$f(x) = a_1 x_1 + a_2 x_2 + b \quad \theta = (a_1, a_2, b) \rightarrow$ 3 parameters

$E(\theta) = \frac{1}{2} \sum_{(x,y) \in X} (a_1 x_1 + a_2 x_2 + b - y)^2$

$\frac{\partial E}{\partial a_1} = \sum (f(x) - y) x_1 \quad \frac{\partial E}{\partial a_2} = \sum (f(x) - y) x_2 \quad \frac{\partial E}{\partial b} = \sum (f(x) - y)$

vetorial notation → $\nabla_\theta E = X^T (X\theta - y)$

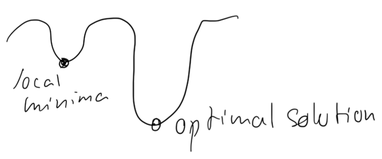
$\theta = \theta - \eta \nabla_\theta E,$

b) The function becomes quadratic, represented by a parabola

Parameters $\theta = (a, b, c)$

$E_\theta = \frac{1}{2} \sum_{(x,y)} (ax + bx^2 + c - y)^2$

Therefor the error surface can have multiple local mimima, so it can get stuck on a non optimal solution



c) η too large can overshoot the optimal point

η too small, takes a lot of iteractions. Taking a long time

$\eta < 0$ increases the error, therefor moving in the wrong direction (getting worse)